Uniform covariance bounds for constrained Ising replicas

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Abstract

We prove that the covariance of finitely many Ising replicas conditioned on fixed magnetizations has a bound independent of the number of replicas, the conserved totals and all replica-dependent external fields, for every fixed interaction matrix. The constraints may conserve the spin totals in each part of an arbitrary partition. This proves the covariance conjecture of Caputo and Morellini. The argument conditions two independent copies on their coordinatewise sum, which cancels the external fields. The remaining disagreements have finitely many symmetric block laws. Pairing identical laws gives a lower bound for the conditioning probability, and a Fourier estimate gives a second-moment numerator of the same order.

1 The covariance question

Magnetization constraints couple otherwise independent replicas of a spin system. A covariance bound uniform in the replica count controls the variance of every linear spin statistic, including in the presence of arbitrary external fields. Such bounds arise in the study of spin-exchange dynamics with conserved magnetizations; see Caputo and Morellini [1, Sections 1.5 and 4.5].

Fix an integer $n \geq 1$, a real symmetric matrix $J \in \mathbb{R}^{n\times n}$, and a partition $\mathcal{A}$ of $[n] = \{1, \dots, n\}$. A configuration of $N \geq 2$ replicas is $\sigma = (\sigma(1), \dots, \sigma(N)) \in (\{-1, 1\}^n)^N$. For integers M_A, $A \in \mathcal{A}$, suppose that

$$\Omega_{N,M} = \left\{ \sigma : \sum_{i=1}^{N} \sum_{j\in A} \sigma_j(i) = M_A \text{ for every } A \in \mathcal{A} \right\}$$

is nonempty. For any $v = (v_{ij}) \in \mathbb{R}^{Nn}$, define

$$\nu_{N,M,v}(\sigma) = \frac{\mathbf{1}_{\Omega_{N,M}}(\sigma)}{Z_{N,M,v}} \exp\left\{ \frac{1}{2} \sum_{i=1}^{N} \sigma(i)^T J \sigma(i) + \sum_{i,j} v_{ij}\sigma_j(i) \right\}. \tag{1}$$

The normalizer $Z_{N,M,v}$ is a positive finite sum. This is the canonical replica measure of [1, Definition 1.9, eqs. (1.23)–(1.24)], with an arbitrary linear tilt and the conserved densities written as integer spin totals: $\rho(A) = (1 + M_A/(N|A|))/2$.

Caputo and Morellini ask in [1, Section 4.5] whether its covariance admits a bound uniform in N, the feasible totals and every tilt v, for arbitrary fixed J. Their Proposition 3.6 gives the bound $2/(1-2\lambda_{\max}(J))$ for a single total constraint when J is positive definite and $\lambda_{\max}(J) < 1/2$; Section 3.4 treats the corresponding partitioned constraints at high temperature. We prove the uniform covariance bound without a spectral-smallness assumption.

Theorem 1.1. *For every $n \geq 1$, every real symmetric $J \in \mathbb{R}^{n\times n}$ and every partition $\mathcal{A}$ of $[n]$, there is a finite constant $C(J, \mathcal{A})$ such that*

$$\left\| \mathrm{Cov}_{\nu_{N,M,v}}(\sigma) \right\|_{\mathrm{op}} \leq C(J, \mathcal{A}) \tag{2}$$

for every $N \geq 2$, every feasible integer vector M, and every $v \in \mathbb{R}^{Nn}$. The covariance is that of all Nn spins, and the operator norm is induced by the ordinary Euclidean norm. More explicitly, for every real array a,

$$\operatorname{Var}_{\nu_{N,M,v}}\left(\sum_{i,j} a_{ij}\sigma_j(i)\right) \leq C(J,\mathcal{A})\sum_{i,j} a_{ij}^2. \tag{3}$$

For fixed n, there are finitely many partitions, so taking their maximum also gives a bound depending only on J. The constant may grow with the fixed dimension and interaction; no optimal dependence is sought here. The result proves the covariance conjecture in [1, Section 4.5]. We do not derive entropy-decay estimates in this note.

The external fields can vary freely from one replica to another, so a conditioning estimate for one fixed tilted law does not provide the required uniformity. Instead, take two independent configurations and condition on their coordinatewise sum. Their difference then has no external field and zero conserved totals. Only the subset of differing sites in each replica matters, giving finitely many possible block laws. Pairing blocks with identical laws makes a Fourier integral nonnegative, which yields a uniform lower bound for the zero-total event. A centered second-moment expansion has the same dependence on the active counts, and the two bounds cancel upon conditioning.

Section 2 proves the required estimate for conditioned block sums. Section 3 applies it to the two-copy difference and completes the proof. Empty subsets and parts with no disagreements are included throughout.

2 Conditioned block sums

For a subset $D \subset [n]$, let J_D be the principal submatrix of J and put

$$\mu_D(s) = \frac{\exp(s^T J_D s)}{\sum_{u\in\{-1,1\}^D}\exp(u^T J_D u)}, \qquad s \in \{-1,1\}^D. \tag{4}$$

The empty cube has one element, so $\mu_\varnothing$ is a point mass. We extend its vectors by zero outside D. These measures are symmetric under $s \mapsto -s$. Set

$$d = |\mathcal{A}|, \qquad B = \sum_{j<k}|J_{jk}|, \qquad \delta = e^{-4B}. \tag{5}$$

The diagonal part of $s^T J_D s$ is constant. Its remaining part has oscillation at most $4B$, and hence

$$\mu_D(s) \geq \delta\, 2^{-|D|} \geq p_* := \delta\, 2^{-n} > 0. \tag{6}$$

Let $X_1, \dots, X_N$ be independent with laws $\mu_{D_1}, \dots, \mu_{D_N}$, where the subsets D_i are arbitrary. We condition their sum on the same conserved quantities as in the original problem. Write

$$m_{iA} = |D_i \cap A|, \qquad K_A = \sum_i m_{iA}, \qquad S_{i,A} = \sum_{j\in D_i\cap A} X_{ij}, \qquad S = \sum_i S_i.$$

The event $S = 0$ is possible exactly when each K_A is even. Necessity follows from parity. For sufficiency, assign half the active coordinates in each part sign $+1$ and half sign -1; every such assignment has positive probability by (6).

Proposition 2.1. *There is a finite $\kappa(J,\mathcal{A})$ such that, for every $N \geq 2$, every sequence of subsets D_i for which all K_A are even, and every real array $a = (a_{ij})$,*

$$\mathbb{E}\left[\left(\sum_i \sum_{j\in D_i} a_{ij}X_{ij}\right)^2 \middle| S = 0\right] \leq \kappa(J,\mathcal{A}) \sum_i \sum_{j\in D_i} a_{ij}^2. \tag{7}$$

The conditioning probability can tend to zero with N. We will bound it from below and the corresponding second-moment numerator from above at the same scale, including parts with few active coordinates.

For $t = (t_A)_{A\in\mathcal{A}} \in \mathbb{R}^d$, put $\phi_i(t) = \mathbb{E}e^{\mathrm{i}t\cdot S_i}$. Symmetry makes ϕ_i real. The uniform cube has characteristic function $\prod_A (\cos t_A)^{m_{iA}}$. Subtracting δ times the uniform law from μ_{D_i} leaves a nonnegative measure of mass $1-\delta$, so

$$|\phi_i(t)| \leq 1 - \delta + \delta \prod_A |\cos t_A|^{m_{iA}}.$$

For $c_1,\dots,c_m \in [0,1]$ with $m \leq n$, the product is at most each c_ℓ, giving $\sum_\ell (1-c_\ell) \leq n(1-\prod_\ell c_\ell)$. Together with $1 - |\cos t| \geq \frac{1}{2}\sin^2 t$ and $1 - x \leq e^{-x}$, this proves

$$|\phi_i(t)| \leq \exp\left(-h \sum_A m_{iA} \sin^2 t_A\right), \qquad h = \frac{\delta}{2n}. \tag{8}$$

The formula also holds for an empty subset D_i.

Lemma 2.2. *If all K_A are even, then*

$$\mathbb{P}(S=0) \geq L \prod_A (1+K_A)^{-1/2}, \qquad L = p_*^{2^n} (r/\pi)^d e^{-ndr^2}, \quad r = \frac{1}{2n}. \tag{9}$$

Proof. Pair blocks with identical subsets D_i, leaving at most one block of each type unpaired. There are at most 2^n unpaired blocks. Each pair contributes an even active count to each part. Thus the unpaired counts are even as well, and we can choose one unpaired spin assignment whose totals are all zero. Its probability is at least $p_*^{2^n}$, by (6).

Let S^{p} be the sum of the paired blocks and K_A^{p} their active counts. If $2r_D$ blocks of type D are retained, finite Fourier inversion gives

$$\mathbb{P}(S^{\mathrm{p}} = 0) = (2\pi)^{-d} \int_{[-\pi,\pi]^d} \prod_{D\subset[n]} \phi_D(t)^{2r_D}\, dt. \tag{10}$$

Here ϕ_D is the common characteristic function of blocks of type D. The integrand is nonnegative everywhere, because these functions are real and their exponents are even. We may therefore retain only the box

$$|t_A| \leq \frac{r}{\sqrt{1+K_A^{\mathrm{p}}}}, \qquad A \in \mathcal{A}.$$

For each paired block, the pointwise Cauchy–Schwarz inequality gives

$$\mathbb{E}(t\cdot S_i)^2 \leq n \sum_A m_{iA} t_A^2 \leq n^2 r^2 = \frac{1}{4}.$$

Using $\cos u \ge 1 - u^2/2$ and $\log(1-z) \ge -2z$ for $0 \le z \le 1/2$, we obtain on this box

$$\phi_i(t) \ge 1 - \frac{n}{2}\sum_A m_{iA} t_A^2 \ge \exp\left(-n \sum_A m_{iA} t_A^2\right).$$

The product in (10) is consequently at least $e^{-n\sum_A K_A^{\mathrm{p}} t_A^2} \ge e^{-ndr^2}$ there. Its integral is bounded below by

$$(r/\pi)^d e^{-ndr^2} \prod_A (1 + K_A^{\mathrm{p}})^{-1/2}.$$

The chosen unpaired assignment is independent of the paired blocks. Multiplying its probability by this bound, then using $K_A^{\mathrm{p}} \le K_A$, proves (9). Empty products are one, so the argument includes the case with no pairs. □

Lemma 2.3. *For all even counts K_A and all real arrays a,*

$$\mathbb{E}\left[F^2 \mathbf{1}_{\{S=0\}}\right] \le U E_a \prod_A (1 + K_A)^{-1/2}, \tag{11}$$

where

$$F = \sum_i \sum_{j \in D_i} a_{ij} X_{ij}, \qquad E_a = \sum_i \sum_{j \in D_i} a_{ij}^2,$$

and the following constants depend only on J and $\mathcal{A}$:

$$\gamma = \frac{2\delta}{n\pi^2}, \qquad H = 4\sqrt{\frac{2n}{\delta}}, \qquad U = e^{\delta}\left(n + \frac{2n^2}{\gamma}\right) H^d. \tag{12}$$

Proof. Put $F_i = \sum_{j \in D_i} a_{ij} X_{ij}$, $E_i = \sum_{j \in D_i} a_{ij}^2$, and $Q(t) = \sum_A K_A t_A^2$. Symmetry gives $\mathbb{E}F_i = 0$. Cauchy–Schwarz gives

$$\mathbb{E}F_i^2 \le n E_i, \qquad \mathbb{E}(t \cdot S_i)^2 \le n \sum_A m_{iA} t_A^2.$$

Define $B_i(t) = \mathbb{E}[F_i e^{\mathrm{i}t \cdot S_i}]$. The vanishing mean and $|e^{\mathrm{i}u} - 1| \le |u|$ yield

$$|B_i(t)| = \left|\mathbb{E}[F_i(e^{\mathrm{i}t \cdot S_i} - 1)]\right| \le n\sqrt{E_i}\left(\sum_A m_{iA} t_A^2\right)^{1/2}.$$

Summing and applying Cauchy–Schwarz over blocks, we get

$$\left(\sum_i |B_i(t)|\right)^2 \le n^2 E_a Q(t). \tag{13}$$

Let $V_i(t) = \mathbb{E}[F_i^2 e^{\mathrm{i}t \cdot S_i}]$. Independence gives the exact expansion

$$\begin{aligned}\mathbb{E}[F^2 e^{\mathrm{i}t \cdot S}] = &\sum_i V_i(t) \prod_{k \ne i} \phi_k(t) \\ &+ \sum_{i \ne j} B_i(t) B_j(t) \prod_{k \notin \{i,j\}} \phi_k(t).\end{aligned} \tag{14}$$

Each term omits at most two characteristic factors. For any omitted index set I with $|I| \le 2$, (8) implies

$$\prod_{k \notin I} |\phi_k(t)| \le \exp\left(-h \sum_A K_A \sin^2 t_A + h \sum_{i \in I, A} m_{iA} \sin^2 t_A \right)$$
$$\le e^{2nh} \exp\left(-h \sum_A K_A \sin^2 t_A \right).$$

The omitted blocks contain at most $2n$ active coordinates. On $[-\pi/2, \pi/2]^d$ we have $\sin^2 t_A \ge 4t_A^2/\pi^2$ and $2nh = \delta$, so the last expression is at most $e^{\delta} e^{-\gamma Q(t)}$. Since $|V_i(t)| \le nE_i$ and $\sum_{i \ne j} |B_i B_j| \le (\sum_i |B_i|)^2$, (14) and (13) give

$$\left| \mathbb{E}[F^2 e^{\mathrm{i}t \cdot S}] \right| \le e^{\delta} E_a [n + n^2 Q(t)] e^{-\gamma Q(t)}. \tag{15}$$

Every S_A is an even integer. Hence

$$\mathbb{E}[F^2 \mathbf{1}_{\{S=0\}}] = \pi^{-d} \int_{[-\pi/2, \pi/2]^d} \mathbb{E}[F^2 e^{\mathrm{i}t \cdot S}] \, dt. \tag{16}$$

Both this formula and (10) follow by integrating each exponential monomial: the normalized integral is one for frequency zero and zero for any other allowed integer frequency. Only finite sums occur.

For $q \ge 0$,

$$[n + n^2 q] e^{-\gamma q} \le \left(n + \frac{2n^2}{\gamma} \right) e^{-\gamma q/2}.$$

Write $\alpha = \gamma/2 = \delta/(n\pi^2)$ and recall $H = 4\sqrt{2n/\delta}$. For an integer $k \ge 1$, the substitution $u = \sqrt{\alpha k}\, t$ gives

$$\frac{1}{\pi} \int_{-\pi/2}^{\pi/2} e^{-\alpha k t^2} \, dt \le \frac{4}{\pi \sqrt{\alpha k}} \le \frac{H}{\sqrt{1+k}}.$$

Indeed, the integral of e^{-u^2} over any finite symmetric interval is at most $2 + 2e^{-1} < 4$: use $e^{-u^2} \le e^{-|u|}$ when $|u| \ge 1$. For $k = 0$ the normalized integral is one, which is also bounded by H. Applying these one-dimensional bounds in (16), with (15), proves (11). □

Proof of Proposition 2.1. Divide (11) by the positive lower bound (9). The factors $\prod_A (1+K_A)^{-1/2}$ cancel, and (7) follows with $\kappa(J, \mathcal{A}) = U/L$. □

3 Disagreements of two independent copies

Proof of Theorem 1.1. Let σ, τ be independent with law $\nu_{N,M,v}$, and set

$$c_{ij} = \frac{\sigma_j(i) + \tau_j(i)}{2}, \qquad x_{ij} = \frac{\sigma_j(i) - \tau_j(i)}{2}.$$

Fix a value of c with positive probability. Where $c_{ij} \ne 0$, the coordinate x_{ij} is zero; where $c_{ij} = 0$, it is $+1$ or -1. Put $D_i = \{j : c_{ij} = 0\}$. Because both copies have total M_A,

$$\sum_i \sum_{j \in A} c_{ij} = M_A, \qquad \sum_i \sum_{j \in A} x_{ij} = 0 \quad (A \in \mathcal{A}).$$

Conversely, for this fixed c, the zero totals of x are precisely the conditions that $c+x$ and $c-x$ both belong to $\Omega_{N,M}$.

The pair energy in a block is

$$\begin{aligned}&\tfrac{1}{2}(c_i+x_i)^T J(c_i+x_i)+\tfrac{1}{2}(c_i-x_i)^T J(c_i-x_i)\\&\qquad+v_i^T(c_i+x_i)+v_i^T(c_i-x_i)=c_i^T Jc_i+x_i^T Jx_i+2v_i^T c_i.\end{aligned}$$

The terms involving v and c are constant under this conditioning. The map from an allowed orientation x to the ordered pair $(c+x, c-x)$ is one-to-one. Thus the conditional law of x is exactly the product of the laws (4) for the subsets D_i, conditioned on $S=0$. In particular, all its active counts K_A are even.

For $F_a(\sigma)=\sum_{i,j} a_{ij}\sigma_j(i)$, the independent-copy identity and Proposition 2.1 now give

$$\begin{aligned}\mathrm{Var}_{\nu_{N,M,v}} F_a &= \tfrac{1}{2}\mathbb{E}\big(F_a(\sigma)-F_a(\tau)\big)^2 = 2\mathbb{E}\left(\sum_{i,j} a_{ij}x_{ij}\right)^2\\&\le \frac{2U}{L}\,\mathbb{E}\sum_i\sum_{j\in D_i} a_{ij}^2 \le \frac{2U}{L}\sum_{i,j} a_{ij}^2.\end{aligned}$$

All conditional laws were taken only at supported values of c. The constants $L>0$ and $U<\infty$ depend on $n, J, \mathcal{A}$ alone.

To obtain the operator bound, let $\Sigma=\mathrm{Cov}_{\nu_{N,M,v}}(\sigma)$. For arbitrary real a, b, Cauchy–Schwarz gives

$$|b^T\Sigma a| \le \sqrt{\mathrm{Var}\,F_b\,\mathrm{Var}\,F_a} \le (2U/L)\|a\|_2\|b\|_2.$$

For $\Sigma a\neq 0$, take $b=\Sigma a/\|\Sigma a\|_2$; the case $\Sigma a=0$ is immediate. This proves (2) with $C(J,\mathcal{A})=2U/L$. □

In terms of n, $d=|\mathcal{A}|$ and $\delta=\exp(-4\sum_{j<k}|J_{jk}|)$, one explicit choice is

$$\begin{aligned}C(J,\mathcal{A}) &= 2e^{\delta+d/(4n)}\left(n+\frac{n^3\pi^2}{\delta}\right)\left(\frac{2^n}{\delta}\right)^{2^n}\\&\quad\times\left(8n\pi\sqrt{\frac{2n}{\delta}}\right)^d.\end{aligned}$$

It is finite for every fixed real symmetric J. Its large size reflects the rough estimates above, rather than an asserted sharp dependence on the interaction or dimension.

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References

[1] P. Caputo and M. Morellini, *Kac's Program and Relative Entropy Decay for Nonlinear Spin-Exchange Dynamics*, arXiv:2511.05223v1 (2025). arXiv:2511.05223v1.